

If we generated the $(1 - \alpha)$ using the same method over many random samples, to estimate many population parameters, in the long run, $(1 - \alpha)$ of those intervals will contain the population parameter.

Confidence Interval: Proportions

Assumptions:

- Data obtained by randomisation
- Distribution is $\sim$ Normal ($np(1 - p) \geq 5$)
- For Binomial $\approx$ Normal, $s_{\hat{p}} \approx se$ approximation,
- $(1 - \alpha)$ Confidence Interval:

$$\hat{p} \pm \underbrace{q_{1-\frac{\alpha}{2}}(se)}_{\text{Margin of Error}} \quad s_{\hat{p}} \approx se = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

$$\text{Margin of Error} \leq W \iff n \geq \frac{q_{1-\frac{\alpha}{2}}^2}{W^2} \hat{p}(1-\hat{p})$$

$n \uparrow \Rightarrow MoE \downarrow$
 $(1 - \alpha) \uparrow \Rightarrow MoE \uparrow$
 $p \uparrow \Rightarrow \hat{p} \uparrow \Rightarrow MoE \uparrow$
 n ultimately depends on costs and limitations. Rectify:
 If $p \approx 0 \vee p \approx 1$, $+2successes$, $+2failures$

Confidence Interval: Mean

Assumptions:

- Data obtained by randomisation
- Population Distribution $\sim$ Normal
- Robust, but not to **outliers**:
 - Summary statistics $\bar{X}$ and s_X sensitive to outliers.
 - $\bar{X}$ no longer $\approx \mu_{\bar{X}} = \mu$

Robustness Confidence Interval is robust wrt. the normality assumption $\Rightarrow$ Performs adequately even when assumption is **modestly** violated

$$\underbrace{\bar{X} \pm t_{df=n-1, 1-\frac{\alpha}{2}}(se)}_{\text{Margin of Error}} \quad s_{\bar{X}} \approx se = \frac{s_X}{\sqrt{n}}$$

s_X is a point estimator of σ .
 For $df \geq 30$ ($n > 30$), and $\mu \pm 3\sigma \approx Range(X)$

$$\text{Margin of Error} \leq W \iff n \geq \frac{\sigma^2 q_{1-\frac{\alpha}{2}}^2}{W^2}$$

$$n \uparrow \Rightarrow MoE \downarrow$$

$$(1 - \alpha) \uparrow \Rightarrow MoE \uparrow$$

$$\sigma^2 \uparrow \Rightarrow s^2 \uparrow \Rightarrow MoE \uparrow$$

t-Distribution

Distribution to allow generalisation for small sample sizes but assumes normal distribution

$$t_{df} \quad df \in \mathbb{R}$$

$$\lim_{df \rightarrow \infty} t_{df} = Norm(0, 1)$$

$$t_{df=30} \approx Norm(0, 1)$$

- Bell-shaped
- Slightly thicker tails than normal
- Shows more variability than normal

9 Significance Tests

Assumptions

Certain conditions or assumptions that the test requires, or makes $\downarrow$

Hypotheses

- H_0 : statement that the parameter takes a particular value (Usually no effect)
- H_a : statement that the parameter falls in some alternative range of values. (Usually represents an effect)
- Assumed to be true until sufficient evidence against the hypothesis
 - One/two-sided test ($>$ or $<$ or $\neq$)

Test Statistic

How far the point estimate of the parameter falls from the H_0 value, usually in no. of se

Is a random variable, each sample is an observation
 Distribution under H_0 is the **null distribution**

P-Value

Probability that the test statistic equals, or is more extreme, than the observed. Calculated by assuming H_0 .
 Smaller P-Value $\Rightarrow$ Stronger evidence against H_0

Conclusion

Interpretation of the P-Value, and in context

Significance level α : the number such that we reject H_0 if the P-Value $\leq$

Statistically Significant: The results are statistically significant, if the data provides sufficient evidence to reject H_0 and support H_a

Types of Errors

	Decision	
Reality	Do not reject H_0	Reject H_0
H_0	Correct Conclusion	Type I Error
H_a	Type II Error	Correct Conclusion

P(Type I Error): Significance level α
P(Type II Error): Complex, but inversely related to P(Type I Error). For fixed α , prob. decreases:

- as parameter moves further into H_a , away from H_0
- as sample size increases

Plot: Probability against p_0 for fixed α , n
 Power of a test = $1 - P(\text{Type II Error})$

Misinterpretations

- "Do not reject $H_0 \neq$ "Accept H_0 "
- Small P-Value does not imply confidence interval is far

Significance Test for p

Assumptions

- Categorical Variable
- Data obtained using randomisation
- Sample size is sufficiently large ($np(1 - p) \geq 5$)
- Sample size is small $\Rightarrow$ two-sided test is robust.
- Otherwise null-dist = $Binom(n, p_0)$

Hypotheses:

$H_0 : p = p_0 \quad H_a : p \neq p_0, p > p_0, p < p_0$

Test Statistic:

$$z = \text{score}_{\hat{p}} \text{ supposing } H_0$$

$$z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}}$$

P-Value:

Null Distribution: $Norm(0, 1)$

$$P - Value = \begin{cases} P(Z < z) & \text{left-sided} \\ P(Z > z) & \text{right-sided} \\ 2 \times P(Z > z) & \text{two-sided} \end{cases}$$

Conclusion: If P-Value is $> \alpha$, strong evidence against H_0 . Otherwise, we do not have strong evidence against H_0 .

Significance test for $\bar{X}$

"One Sample t-Test"

Assumptions

- Quantitative Variable
- Data obtained using randomisation
- Population distribution $\approx$ Normal
- Two-sided test is robust (because CLT)
- Except when n is small and H_a is one-sided, sampling distribution is no longer t dist.

Hypotheses

$H_0 : \mu = \mu_0 \quad H_a : \mu \neq \mu_0, \mu > \mu_0, \mu < \mu_0$

Test Statistic $T = \text{Score}_{\bar{X}}$ supposing H_0

$$T = \frac{\bar{X} - \mu_0}{\frac{s}{\sqrt{n}}}$$

P-Value:

Null Distribution: $t_{df=n-1}$

$$P - Value = \begin{cases} P(t_{df} < T) & \text{left-sided} \\ P(t_{df} > T) & \text{right-sided} \\ 2 \times P(t_{df} > T) & \text{two-sided} \end{cases}$$

Conclusion If P-Value is $> \alpha$, strong evidence against H_0 . Otherwise, we do not have strong evidence against H_0 .

Two-sided Test vs. Confidence Interval

Two-sided test P-Value $\leq \alpha \iff (1 - \alpha)$ Conf-Int does not contain H_0 value

10 Bivariate Inference Methods

Sampling Distribution of $(p_1 - p_2)$

$$\mu_{\hat{p}_1 - \hat{p}_2} = p_1 - p_2$$

$$s_{\hat{p}_1 - \hat{p}_2} = \sqrt{\frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}}$$

Confidence Interval for $(p_1 - p_2)$

Assumptions

- Categorical Response variable observed
- Independent random samples for the two groups
- Large sample sizes, $np > 10 \wedge n(1 - p) > 10$ for each group

$$(1 - \alpha) \text{ Confidence Interval:}$$

$$\hat{p}_1 - \hat{p}_2 \pm q_{1-\frac{\alpha}{2}}(se)$$

$$se = \sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$$

If confidence interval contains 0, it is plausible that $(p_1 - p_2) = 0$, and the proportions might be equal.
 Sign of values: $p_1 > p_2$ or $p_1 < p_2$
 Magnitude of values: The size of the true difference in proportions

Significance Test for $(p_1 - p_2)$

Same **assumptions** as confidence interval

Hypotheses

$H_0 : p_1 = p_2, H_a : p_1 \neq p_2, p_1 > p_2, p_1 < p_2$

Test Statistic, $\hat{p}$ is the pooled estimate

$$z = \frac{(\hat{p}_1 - \hat{p}_2) - 0}{se_0}, se_0 = \sqrt{\hat{p}(1-\hat{p})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}$$

P-Value

Null Distribution: $Norm(0, 1)$

Conclusion Groups are (statistically) significantly different if P-Value is small

Sampling Distribution of $(\mu_1 - \mu_2)$

$$\mu_{\bar{X}_1 - \bar{X}_2} = \mu_1 - \mu_2 \quad s_{\bar{X}_1 - \bar{X}_2} = \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$$

Null Distribution: $t_{df} . df$ is complex.

Confidence Interval for $(\mu_1 - \mu_2)$

Assumptions

- Quantitative response variable observed
- Independent random samples for the two groups
- Approximately normal population dist. for each group
- Robust, except to outliers [Confidence Interval: Mean]

$$(1 - \alpha) \text{ Confidence Interval:}$$

$$(\bar{X}_1 - \bar{X}_2) \pm t_{df, 1-\frac{\alpha}{2}}(se), se = \sqrt{\frac{s_1^2}{n_1} + \frac{s_1^2}{n_2}}$$

Significance Test for $(\mu_1 - \mu_2)$

Same **assumptions** as confidence interval

Hypotheses

$H_0 : \mu_1 = \mu_2, H_a : \mu_1 \neq \mu_2 \mu_1 > \mu_2 \mu_1 < \mu_2$

Test Statistic, se as confidence interval

$$t = \frac{(\bar{X}_1 - \bar{X}_2) - 0}{se_0}$$

P-Value Null Distribution: t_{df}

Conclusion Groups are (statistically) significantly different if P-Value is small

Significance Test for $(\mu_1 - \mu_2), \sigma_1 = \sigma_2$

F test for comparing standard deviation:

P-Value $< \alpha = 0.05$

$\uparrow$ NOT robust to population normality assumption

Assumptions, **Test Statistic**, **P-Value**, **Conclusion** are all the same as for non-equal std. dev, but with:

$$se = s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$$

$$s_p = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{x_1 + n_2 - 2}}$$

$$df = n_1 + n_2 - 2$$

11 Linear Regression

$$Y = \bar{Y} - b(\bar{X}) + R\left(\frac{s_Y}{s_X}\right)X + \epsilon$$

Y-intercept $\bar{Y} - b\bar{X}$

Predicted value of y when $x = 0$, Might have no interpretative value (if no observations near $x=0$).

Slope $R\left(\frac{s_Y}{s_X}\right)$

Same sign as R. Amount that $\hat{y}$ changes with one unit increase of x R^2 is the % of variability in the response variable that can be explained by the linear relationship with the explanatory variable.

Error term ϵ

Assumptions

- Data obtained by randomisation
- Relationship between X and Y is linear
- Error term $\epsilon \sim Norm(0, \sigma^2)$ where σ is a constant

Implications of Assumptions

$\forall X (Y \sim Norm(\beta_0 + \beta_1 X, \sigma^2))$

Ordinary Least Squares Estimation

Best fit regression is the minimisation of $\sum_{i=0}^{n-1} e_i^2$

Interpreting Info of Linear Regression Models

$$\hat{Y} = \hat{\beta}_0 - \hat{\beta}_1 X$$

Residuals

Quartiles of the residuals of each point with the model

Estimate $\hat{\beta}_i$

$\Rightarrow$ Point Estimates of each coefficient in the model

Std. Error

Standard error of each coefficient. Can be used to obtain a confidence interval

Residual Standard Error

The standard error of $\hat{\sigma}$ For each point $x_i, e_i = y_i - \hat{y}_i$

- Could be normalised, to get standard residuals $SR \sim Norm(0, 1)$
- σ is the measure of how far the observations can deviate from best-fit line
- σ^2 is the measure of how far the observations can deviate from the best-fit line

Multiple R-squared

Coefficient of Determination of linear model

Hypothesis Testing on Linear Models

t-Test

The significance of one regressor.

Assumptions Same as assumptions of model

Hypothesis

$H_0 : \beta_i = 0$ OR Regressor i is NOT significant

$H_a : \beta_i \neq 0$ OR Regressor i is significant

Test Statistic

$$t = \frac{\hat{\beta}_i}{se}$$

P-Value

Null-Distribution: t Distribution, $df = n - \text{no. of coefficients}$

Conclusion

The coefficient is (not) significantly different from 0 at α -level

F-test

Assumptions Same as assumptions of model

Hypothesis

H_0 : model is NOT significant OR all the coefficients except β_0 are zero

H_a : model is significant OR at least one of the coefficients except β_0 are non-zero

Test Statistic F-statistic from R output
 $F = t^2$ for Simple Linear Regression

P-Value

Null-Distribution: F Distribution,
 $df1$ =no. of coefficients $- 1$
 $df2$ = n -no. of coefficients

Find right-sided P-Value on F Distribution

Conclusion

The data provides (in)sufficient evidence that the built model is significant.

$P - Value < \alpha \Rightarrow$ ALL regressors used in the model are not significant, $Y = \beta_0$

Checking Assumptions of Linear Model

Before fitting model, scatterplot of Y against X:

- Linearity (No curved bands)
- Constant Variance (Funnel shape)

To verify the assumption $\epsilon \sim Norm(0, \sigma^2)$

- SR against $\hat{Y}_i$
- SR against X
 - Points scatter randomly around 0, within $(-3, 3)$
 - Funnel shaped observed $\Rightarrow$ Constant variance assumption violated
- Histogram of SR
- QQ Plot of SR
 - SR has a normal distribution
 - Skewed Distribution $\Rightarrow$ Normality assumption violated

Possible Fixes

- Add higher order terms
- Transform response into $\ln(Y)$, $\sqrt{Y}$, $\frac{1}{Y}$
- Add more regressors
- Non-linear model required
- $|SR| > 3 \Rightarrow$ Potential outlier
- Cook's Distance $> 1 \Rightarrow$ Potential influential point
- Avoid Extrapolation (Estimation using regressors outside domain)

Coefficient of Determination

Interpretation:

The proportion of total variation of the response (of sample mean $\bar{Y}$) that is explained by the model.

For simple model: $\sqrt{R^2} = |Cor(X, Y)|$
 $R^2 = 1 \Rightarrow \forall i (Y_i = Y_i)$ Adding regressors will always increase, or not change R^2 . Use adjusted R^2

$$R_{adj}^2 = 1 - \frac{(1 - R^2)(n - 1)}{n - \text{no. of coefficients}}$$

Indicator Variables

Each indicator splits the model into two equations, on whether the indicator is 1 or 0

n categories require $n - 1$ indicators
 Identify the reference category when every indicator is 0.
 Use anova P-Value for significance of categorical variables